

Problem Session 2

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Doing a Phonological Analysis

Steps:

- ▶ examine the distribution of sounds within the given data (for each sound in question, determine the set of phonetic environments in which it can occur).
- ▶ to begin a phonemic analysis, first look for minimal pairs.
- ▶ if there are no minimal pairs and the environments do not overlap, then the two sounds can be in complementary distribution.
- ▶ then, determine an underlying phoneme and come up with a generalizable rule that explains under which conditions one segment becomes another.
- ▶ to come up with a rule, try to find natural classes.

Brief Exercise

- ▶ What are minimal pairs? Give examples.
- ▶ Do all languages have the same phonemic inventory?
- ▶ What is the difference between complementary distribution and contrastive distribution?
- ▶ Which one is more abstract? Phonetic realization sounds or phonemes? Why?

Korean

Korean is a “language isolate,” meaning that it is not linguistically related to other languages. It is spoken in Korea. In the following Korean words, you will find the sounds [s] and [ʃ]. Determine whether the sounds [s] and [ʃ] are allophones of the same phoneme or separate phonemes. If the sounds are allophones of the same phoneme, give the basic and derived allophones and the environment in which the derived allophone occurs.

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|------------------|----------------|
| a. [ʃi] | ‘poem’ |
| b. [miʃin] | ‘superstition’ |
| c. [ʃinmun] | ‘newspaper’ |
| d. [tʰaksanʃige] | ‘table clock’ |
| e. [ʃilsu] | ‘mistake’ |
| f. [oʃip] | ‘fifty’ |
| g. [paŋʃik] | ‘method’ |
| h. [kanʃik] | ‘snack’ |
| i. [kaʃi] | ‘thorn’ |
| j. [sal] | ‘flesh’ |
| k. [kasu] | ‘singer’ |
| l. [sanmun] | ‘prose’ |
| m. [kasəl] | ‘hypothesis’ |
| n. [miso] | ‘smile’ |
| o. [susek] | ‘search’ |
| p. [tapsa] | ‘exploration’ |
| q. [so] | ‘cow’ |

Sindhi

The following data are from Sindhi, an Indo-European language of the Indo-Aryan family, spoken in Pakistan and India. Examine the distribution of the phones [p], [p^h], and [b]. Determine if the three are allophones of separate phonemes or allophones of the same phoneme. What is your evidence? Is the relationship among the sounds the same as in English? Why or why not? [ŋ] and [d] are the voiced retroflex nasal and oral stop.

- | | |
|-------------------------|-----------------|
| a. [pənθ] | ‘leaf’ |
| b. [dʌk ^h ə] | ‘grape’ |
| c. [dəpθ] | ‘fear’ |
| d. [p ^h əŋu] | ‘hood of snake’ |
| e. [kənθ] | ‘ear’ |
| f. [perθ] | ‘foot’ |
| g. [barθ] | ‘weight’ |
| h. [k ^h irθ] | ‘milk’ |
| i. [put ^h i] | ‘back’ |
| j. [bənθ] | ‘forest’ |
| k. [p ^h ərθ] | ‘arrow head’ |
| l. [abθ] | ‘water’ |

Mokilese

Mokilese is an Austronesian language, spoken in Micronesia. Examine the distribution of the voiced and voiceless vowel pairs: [i, i̥] and [u, u̥] (voiceless vowels have a circle under the phonetic vowel symbol). For each pair, determine whether they are allophones of different phonemes or allophones of the same phoneme. Provide evidence for your answer. If they are allophones of one phoneme, state the contexts in which each sound occurs and decide which sound is the basic sound.

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|-----------------|-----------------------|
| a. [pi̯senmon] | ‘hair’ |
| b. [tu̯pukda] | ‘bought’ |
| c. [pu̯ko] | ‘basket’ |
| d. [pi̯tʃumene] | ‘this pig’ |
| e. [su̯pwo] | ‘firewood’ |
| f. [si̯ko] | ‘to talk about’ |
| g. [uduk] | ‘flesh’ |
| h. [masak] | ‘hard’ |
| i. [li̯ŋ] | ‘beautiful’ |
| j. [pilda] | ‘to choose’ |
| k. [poki] | ‘to strike something’ |
| l. [lu̯dʒ] | ‘to jump’ |